

Lower bound reasoning with eHL

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The eHL judgment. Let Mem denote the set of program memories and let $\text{Exp} = \text{Mem} \rightarrow \mathbb{R}^+$ be the set of non-negative extended-real expectations. For predicates $P, Q : \text{Mem} \rightarrow \mathbb{B}$, expectations $f, g \in \text{Exp}$, and a program C , the eHL judgment

$$\{P \mid f\} \bowtie C \{Q \mid g\} \quad \bowtie \in \{\leq, \geq\}$$

is defined by:

$$\{P \mid f\} \leq C \{Q \mid g\} \triangleq \forall m \in \text{Mem}. P(m) \implies f(m) \leq \mathbb{E}_{C,m}[[Q] \cdot g] \quad (1)$$

$$\{P \mid f\} \geq C \{Q \mid g\} \triangleq \forall m \in \text{Mem}. P(m) \implies f(m) \geq \mathbb{E}_{C,m}[[Q] \cdot g] \quad (2)$$

where $\mathbb{E}_{C,m}[[Q] \cdot g]$ denotes the expected value of $[Q] \cdot g$ over the output distribution of C starting from memory m , or equivalently $\text{wp}[C]([Q] \cdot g)(m)$. In both cases, the precondition P restricts the obligation to memories where P holds; at memories where P is false the judgment is vacuously satisfied.

Validity. Let C_m denote the output distribution of C from memory m . A judgment $\{P \mid f\} \bowtie C \{Q \mid g\}$ is *valid* if both of the following hold:

- (1) $\forall m \in \text{Mem}. P(m) \implies \text{supp}(C_m) \subseteq Q$
- (2) $\forall m \in \text{Mem}. P(m) \implies f(m) \bowtie \mathbb{E}_{C,m}[g]$

Condition (1) requires that all output memories satisfy the postcondition Q . Condition (2) requires that $f(m)$ bounds the expected value of g under the output distribution of C .

$$\begin{array}{c} \overline{\{P \mid f\} \bowtie \text{skip} \{P \mid f\}} \text{ [SKIP]} \\ \overline{\{P[x/E] \mid f[x/E]\} \bowtie x \leftarrow E \{P \mid f\}} \text{ [ASSIGN]} \\ \overline{\{\forall v \in \text{supp}(S), P[x/v] \mid \mathbb{E}_S[\lambda v. f[x/v]]\} \bowtie x \xleftarrow{\$} S \{P \mid f\}} \text{ [SAMPLE]} \\ \overline{\frac{\{P \mid f\} \bowtie C \{R \mid h\} \quad \{R \mid h\} \bowtie D \{Q \mid g\}}{\{P \mid f\} \bowtie C; D \{Q \mid g\}}} \text{ [SEQ]} \\ \overline{\frac{\{B \wedge P \mid f\} \bowtie C \{Q \mid g\} \quad \{\neg B \wedge P \mid f\} \bowtie D \{Q \mid g\}}{\{P \mid f\} \bowtie \text{if } B \text{ then } C \text{ else } D \{Q \mid g\}}} \text{ [IF]} \\ \overline{\frac{B \wedge P \models f_0 \bowtie 0 \quad \forall k. \{B \wedge P \mid f_{k+1}\} \bowtie C \{P \mid f_k\} \quad \neg B \wedge P \models f_k \bowtie g}{\{P \mid \text{lub} f_k\} \bowtie \text{while } B \text{ do } C \{ \neg B \wedge P \mid g\}}} \text{ [WHILE]} \end{array}$$

Fig. 1. eHL for lower and upper-bounds. Here $\bowtie \in \{\leq, \geq\}$.

Notation. For a predicate $P : \text{Mem} \rightarrow \mathbb{B}$ we use two indicator-style operators, one tailored to each direction of reasoning:

$$[P](m) = \begin{cases} 1 & \text{if } P(m) \\ 0 & \text{otherwise} \end{cases} \quad (\text{used for lower bounds})$$

$$\langle P \rangle(m) = \begin{cases} 1 & \text{if } P(m) \\ \infty & \text{otherwise} \end{cases} \quad (\text{used for upper bounds})$$

The witness sequence $\{r_i\}_{i \in \mathbb{N}}$. The rule of Figure 2 uses a sequence $\{r_i\}_{i \geq 0}$ of non-negative expectations as a termination witness: intuitively, r_i is a (user-chosen) upper bound on how much of the post-expectation mass of f is still trapped inside the loop after i body executions. Premise 2 ties r_i to f at $i = 0$ by requiring $r_0 \geq f$; Premise 3 propagates the bound across one iteration of the body; and Premise 4 forces the bound — and thus the trapped mass — to vanish in the limit.

$$\frac{\{B \wedge P \mid f\} \leq C \{P \mid f\} \quad B \wedge P \models r_0 \geq f \quad \forall i \geq 0. \{B \wedge P \mid r_{i+1}\} \geq C \{\top \mid r_i\} \quad B \wedge P \models \lim_{i \rightarrow \infty} r_i = 0}{\{P \mid f\} \leq \text{while } B \text{ do } C \{ \neg B \wedge P \mid f \}} \quad [\text{WHILE}]$$

Fig. 2. The WHILE rule for eHL lower bounds. The rule is specific to the lower-bound judgment: Premise 1 is a lower-bound triple on the body, while Premise 3 is an *upper*-bound triple used purely to bound the in-loop mass. The sequence $\{r_i\}_{i \geq 0}$ consists of non-negative expectations $r_i : \text{State} \rightarrow \mathbb{R}^+$.

A Soundness of the WHILE Rule

Goal. We want to show that the following rule is sound:

$$[P] \cdot f \leq \text{wp} [\text{while } B \text{ do } C] ([P \wedge \neg B] \cdot f)$$

Definitions. Define the family of approximating programs:

$$W_n \triangleq (\text{if } B \text{ then } C)^n ; \text{if } B \text{ then abort}$$

where W_n allows at most n iterations; if the loop would continue beyond n , it aborts. Define:

$$I_n \triangleq [P] \cdot \text{wp} [W_n] ([P \wedge \neg B] \cdot f)$$

LEMMA A.1 (RECURSIVE CHARACTERISATION OF I_n).

$$I_0 = [P \wedge \neg B] \cdot f \quad \text{and} \quad I_{n+1} = [P \wedge B] \cdot \text{wp} [C] ([P] \cdot I_n) + [P \wedge \neg B] \cdot f$$

PROOF. By induction on n .

Base case $n = 0$.

$$\begin{aligned} I_0 &= [P] \cdot \text{wp} [\text{if } B \text{ then abort}] ([P \wedge \neg B] \cdot f) \\ &= [P] \cdot ([B] \cdot \text{wp} [\text{abort}] ([P \wedge \neg B] \cdot f) + [\neg B] \cdot [P \wedge \neg B] \cdot f) \\ &= [P] \cdot ([B] \cdot 0 + [\neg B] \cdot [P \wedge \neg B] \cdot f) \\ &= [P] \cdot [P \wedge \neg B] \cdot f = [P \wedge \neg B] \cdot f \quad \checkmark \end{aligned}$$

where we used $\text{wp} [\text{abort}] (\cdot) = 0$ and $[P] \cdot [P \wedge \neg B] = [P \wedge \neg B]$.

Inductive step $n \rightarrow n + 1$.

$$I_{n+1} = [P] \cdot \text{wp} [(\text{if } B \text{ then } C)^{n+1} ; \text{if } B \text{ then abort}] ([P \wedge \neg B] \cdot f)$$

Unfolding $(\text{if } B \text{ then } C)^{n+1} = \text{if } B \text{ then } C ; (\text{if } B \text{ then } C)^n$ and applying the SEQ rule:

$$= [P] \cdot \text{wp} [\text{if } B \text{ then } C] (\text{wp} [W_n] ([P \wedge \neg B] \cdot f))$$

By sub-induction one verifies $[P] \cdot I_k = I_k$ for all k (since $I_0 = [P \wedge \neg B] \cdot f$ and each I_{k+1} carries $[P \wedge B]$ or $[P \wedge \neg B]$ as a factor), so in particular:

$$\text{wp} [W_n] ([P \wedge \neg B] \cdot f) = [P] \cdot \text{wp} [W_n] ([P \wedge \neg B] \cdot f) = I_n$$

Therefore:

$$\begin{aligned} &= [P] \cdot \text{wp} [\text{if } B \text{ then } C] (I_n) \\ &= [P] \cdot ([B] \cdot \text{wp} [C] (I_n) + [\neg B] \cdot I_n) \\ &= [P \wedge B] \cdot \text{wp} [C] (I_n) + [P \wedge \neg B] \cdot I_n \end{aligned}$$

Using $[P] \cdot I_n = I_n$ and the identity $[P \wedge \neg B] \cdot I_n = [P \wedge \neg B] \cdot f$ (which follows from $[\neg B] \cdot I_n = [\neg B] \cdot [P \wedge \neg B] \cdot f = [P \wedge \neg B] \cdot f$):

$$I_{n+1} = [P \wedge B] \cdot \text{wp} [C] ([P] \cdot I_n) + [P \wedge \neg B] \cdot f \quad \checkmark$$

□

LEMMA A.2 (KEY INEQUALITY). Assuming premises (a), (b), (c), for all $n \geq 0$:

$$[P] \cdot I_n + [P] \cdot r_n \geq [P] \cdot f$$

PROOF. By induction on n .

Base case $n = 0$.

$$[P] \cdot I_0 + [P] \cdot r_0 = [P \wedge \neg B] \cdot f + [P] \cdot r_0$$

By premise (b), $[P] \cdot r_0 \geq [P \wedge B] \cdot f$, so:

$$\geq [P \wedge \neg B] \cdot f + [P \wedge B] \cdot f = [P] \cdot f \quad \checkmark$$

Inductive step $n \rightarrow n + 1$. Expanding I_{n+1} by Theorem A.1:

$$[P] \cdot I_{n+1} + [P] \cdot r_{n+1} = [P \wedge B] \cdot \text{wp}[C]([P] \cdot I_n) + [P \wedge \neg B] \cdot f + [P] \cdot r_{n+1}$$

By premise (c), $[P] \cdot r_{n+1} \geq [P \wedge B] \cdot \text{wp}[C]([P] \cdot r_n)$:

$$\geq [P \wedge B] \cdot \text{wp}[C]([P] \cdot I_n) + [P \wedge B] \cdot \text{wp}[C]([P] \cdot r_n) + [P \wedge \neg B] \cdot f$$

By linearity of $\text{wp}[C](\cdot)$:

$$= [P \wedge B] \cdot \text{wp}[C]([P] \cdot I_n + [P] \cdot r_n) + [P \wedge \neg B] \cdot f$$

By the induction hypothesis $[P] \cdot I_n + [P] \cdot r_n \geq [P] \cdot f$ and monotonicity of $\text{wp}[C](\cdot)$:

$$\geq [P \wedge B] \cdot \text{wp}[C]([P] \cdot f) + [P \wedge \neg B] \cdot f$$

By premise (a), $[P \wedge B] \cdot \text{wp}[C]([P] \cdot f) \geq [P \wedge B] \cdot f$:

$$\geq [P \wedge B] \cdot f + [P \wedge \neg B] \cdot f = [P] \cdot f \quad \checkmark$$

□

Main result. From Theorem A.2, for all $n \geq 0$:

$$[P] \cdot I_n + [P] \cdot r_n \geq [P] \cdot f$$

Taking the limit:

$$\lim_n([P] \cdot I_n) + \lim_n([P] \cdot r_n) \geq [P] \cdot f$$

By condition (d), $\lim_n r_n = 0$, so $\lim_n([P] \cdot r_n) = 0$:

$$\lim_n([P] \cdot I_n) \geq [P] \cdot f$$

By Scott-continuity and Theorem A.1, $\lim_n I_n = \text{wp}[\text{while } B \text{ do } C]([P \wedge \neg B] \cdot f)$, giving:

$$\text{wp}[\text{while } B \text{ do } C]([P \wedge \neg B] \cdot f) \geq [P] \cdot f$$

which is exactly the desired conclusion. □

B Completeness

Theorem: Completeness. Assumptions:

1. $[P] \cdot f \leq \text{wp}[\text{while } B \text{ do } C]([\neg B \wedge P] \cdot f)$
2. $[B \wedge P] \cdot f \leq \text{wp}[C]([P] \cdot f)$

Definitions:

$$H \triangleq \text{wp}[\text{while } B \text{ do } C]([\neg B \wedge P] \cdot f)$$

$$I_0 \triangleq [\neg B \wedge P] \cdot f$$

$$I_{i+1} \triangleq [B] \cdot \text{wp}[C](I_i) + I_0$$

$$r_i \triangleq H - I_i$$

Verification of Premise 2:

$$\begin{aligned}
& B \wedge P \implies I_0 = 0 \\
& r_0 = H - I_0 = H \\
& P \text{ holds so: } H \geq [P] \cdot f \implies H \geq f \\
& [B \wedge P] \cdot r_0 \geq [B \wedge P] \cdot f
\end{aligned}$$

Verification of Premise 3:

$$\begin{aligned}
& H = [B] \cdot \text{wp } [C] (H) + I_0 \\
& r_{i+1} = H - I_{i+1} \\
& r_{i+1} = ([B] \cdot \text{wp } [C] (H) + I_0) - ([B] \cdot \text{wp } [C] (I_i) + I_0) \\
& r_{i+1} = [B] \cdot (\text{wp } [C] (H) - \text{wp } [C] (I_i)) \\
& r_{i+1} = [B] \cdot \text{wp } [C] (H - I_i) = [B] \cdot \text{wp } [C] (r_i) \\
& \text{By Monotonicity: } r_i \geq [P] \cdot r_i \implies \text{wp } [C] (r_i) \geq \text{wp } [C] ([P] \cdot r_i) \\
& [B \wedge P] \cdot r_{i+1} \geq [B \wedge P] \cdot \text{wp } [C] ([P] \cdot r_i)
\end{aligned}$$

Verification of Premise 4:

$$\begin{aligned}
& \lim_{i \rightarrow \infty} I_i = H \\
& \lim_{i \rightarrow \infty} r_i = H - \lim_{i \rightarrow \infty} I_i = H - H = 0 \\
& \lim_{i \rightarrow \infty} r_i = 0
\end{aligned}$$

C Derived Rule

Multiplicative decay.

$$\frac{\{B \wedge P \mid f\} \leq C \{P \mid f\} \quad B \wedge P \models V \geq f \quad \{B \wedge P \mid \beta \cdot V\} \geq C \{\top \mid V\} \quad \beta \in [0, 1] \quad B \wedge P \models V < \infty}{\{P \mid f\} \leq \text{while } B \text{ do } C \{ \neg B \wedge P \mid f \}} \quad \text{[WHILE-MULT]}$$

C.1 Proof via Main Rule Soundness.**Premises:**

$$\begin{aligned}
(1) & [B \wedge P] \cdot f < \text{wp } [C] ([P] \cdot f) \\
(2) & V \geq f \\
(3) & \text{wp } [C] (V) < \beta \cdot V, \quad \beta < 1 \\
(4) & V < \infty
\end{aligned}$$

Proof Strategy: $r_i \triangleq \beta^i \cdot V$.

Verification of Main Rule Premise 2:

$$r_0 = V \xrightarrow{\text{by (2)}} r_0 \geq f \implies [B \wedge P] \cdot r_0 \geq [B \wedge P] \cdot f$$

Verification of Main Rule Premise 3:

$$\begin{aligned}
\text{wp } [C] (r_i) &= \text{wp } [C] (\beta^i \cdot V) = \beta^i \cdot \text{wp } [C] (V) \\
&< \beta^i \cdot (\beta \cdot V) \quad (\text{by (3)}) \\
&= \beta^{i+1} \cdot V = r_{i+1} \\
\implies [B \wedge P] \cdot r_{i+1} &\geq [B \wedge P] \cdot \text{wp } [C] (r_i)
\end{aligned}$$

Verification of Main Rule Premise 4:

$$\lim_{i \rightarrow \infty} r_i = \lim_{i \rightarrow \infty} (\beta^i \cdot V) \xrightarrow{\beta < 1, V < \infty} 0 \implies \lim_{i \rightarrow \infty} [B \wedge P] \cdot r_i = 0$$

Conclusion: Soundness follows from the main rule. $\checkmark$

C.2 Direct Proof.

Definitions: (Recall I_n from Theorem A.1)

$$r_i \triangleq \beta^i \cdot V$$

Auxiliary inequality on $B \wedge P$:

$$\begin{aligned}
\text{wp } [C] (V) &\leq \beta \cdot V \quad (\text{premise 3}) \\
\text{wp } [C] (r_n) &= \beta^n \cdot \text{wp } [C] (V) \leq \beta^{n+1} \cdot V = r_{n+1} \quad (\text{by linearity}) \\
\text{wp } [C] ([P] \cdot r_n) &\leq \text{wp } [C] (r_n) \leq r_{n+1} \quad (\text{by monotonicity, } [P] \cdot r_n \leq r_n) \\
[B \wedge P] \cdot r_{n+1} &\geq [B \wedge P] \cdot \text{wp } [C] ([P] \cdot r_n)
\end{aligned}$$

Key inequality: $\forall n. [P] \cdot I_n + [P] \cdot r_n \geq [P] \cdot f$ (by induction)

Base case $n = 0$:

$$\begin{aligned}
[P] \cdot I_0 + [P] \cdot r_0 &= [\neg B \wedge P] \cdot f + [P] \cdot V \\
&\geq [\neg B \wedge P] \cdot f + [B \wedge P] \cdot V \\
&\geq [\neg B \wedge P] \cdot f + [B \wedge P] \cdot f \quad (\text{by premise 2}) \\
&= [P] \cdot f
\end{aligned}$$

Inductive step $n \rightarrow n + 1$ (using Theorem A.1 for I_{n+1}):

$$\begin{aligned}
[P] \cdot I_{n+1} + [P] \cdot r_{n+1} &= [B \wedge P] \cdot \text{wp } [C] ([P] \cdot I_n) + [\neg B \wedge P] \cdot f + [P] \cdot r_{n+1} \\
&\geq [B \wedge P] \cdot \text{wp } [C] ([P] \cdot I_n) + [\neg B \wedge P] \cdot f + [B \wedge P] \cdot \text{wp } [C] ([P] \cdot r_n) \\
&\quad \text{since } [P] \cdot r_{n+1} \geq [B \wedge P] \cdot r_{n+1} \geq [B \wedge P] \cdot \text{wp } [C] ([P] \cdot r_n) \\
&\quad (\text{by } [P] \geq [B \wedge P] \text{ and auxiliary inequality}) \\
&= [B \wedge P] \cdot \text{wp } [C] ([P] \cdot I_n + [P] \cdot r_n) + [\neg B \wedge P] \cdot f \quad (\text{by linearity}) \\
&\geq [B \wedge P] \cdot \text{wp } [C] ([P] \cdot f) + [\neg B \wedge P] \cdot f \quad (\text{by IH + monotonicity}) \\
&\geq [B \wedge P] \cdot f + [\neg B \wedge P] \cdot f \quad (\text{by premise 1}) \\
&= [P] \cdot f
\end{aligned}$$

Limit argument:

$$\begin{aligned}
\beta \in [0, 1), B \wedge P &\implies V < \infty \implies \lim_{n \rightarrow \infty} r_n = 0 \\
\lim_{n \rightarrow \infty} I_n &= \text{wp } [\text{while } B \text{ do } C] ([\neg B \wedge P] \cdot f) \quad (\text{by Scott-continuity}) \\
\implies [P] \cdot f &\leq \text{wp } [\text{while } B \text{ do } C] ([\neg B \wedge P] \cdot f) \quad \checkmark
\end{aligned}$$

C.3 Example: Geometric Termination

Program.

$$C_{\text{geo}} \triangleq \text{while } b \text{ } b := \text{Ber}(p); \ c := c + 1$$

At each iteration the loop resamples $b \sim \text{Ber}(p)$ and increments the counter c . The loop terminates when $\neg b$, which occurs geometrically with $\mathbb{E}[T] = \frac{1}{1-p}$.

Instantiation. Set $B \equiv (b)$, $P \equiv \text{true}$,

$$f(b, c) \triangleq c + \frac{[b]}{1-p}, \quad V(b, c) \triangleq \left(c + \frac{1}{1-p}\right) \cdot [b], \quad \beta \triangleq p(2-p).$$

Note that $V = f$ on $B \wedge P$ (i.e. at b) and $V = 0$ otherwise.

Verification of the premises. **Premise 1** ($\{B \wedge P \mid f\} \leq C \{P \mid f\}$):

$$\begin{aligned} \text{wp}[C_{\text{body}}](f)(\text{true}, c) &= p f(\text{true}, c+1) + (1-p) f(\text{false}, c+1) \\ &= p \left(c + 1 + \frac{1}{1-p}\right) + (1-p)(c+1) = c + \frac{1}{1-p} = f(\text{true}, c). \checkmark \end{aligned}$$

Premise 2 ($V \geq f$ on $B \wedge P$): $V(\text{true}, c) = c + \frac{1}{1-p} = f(\text{true}, c)$. $\checkmark$

Premise 3 ($\text{wp}[C_{\text{body}}](V) \leq \beta \cdot V$ on $B \wedge P$):

$$\begin{aligned} \text{wp}[C_{\text{body}}](V)(\text{true}, c) &= p \cdot V(\text{true}, c+1) + (1-p) \cdot V(\text{false}, c+1) \\ &= p \left(c + 1 + \frac{1}{1-p}\right) = p V(\text{true}, c) + p. \end{aligned}$$

We need $pV + p \leq p(2-p)V$, i.e. $p \leq p(1-p)V = p(1-p)\left(c + \frac{1}{1-p}\right) = p((1-p)c + 1)$, which reduces to $0 \leq (1-p)c$, true for all $c \geq 0$ (with equality at $c = 0$). $\checkmark$

Premise 4 ($\beta \in [0, 1)$): $p(2-p) < 1 \iff (1-p)^2 > 0$, which holds for $p \in (0, 1)$. $\checkmark$

Premise 5 ($V < \infty$ on $B \wedge P$): $V(\text{true}, c) = c + \frac{1}{1-p} < \infty$ for any fixed c . $\checkmark$

Witnesses and conclusion. The rule instantiates with $r_i(\text{true}, c) = (p(2-p))^i \left(c + \frac{1}{1-p}\right)$, giving

$$\vdash \{ \text{true} \mid c + \frac{1}{1-p} \} \leq C_{\text{geo}} \{ \neg b \mid c \}.$$

This certifies $\mathbb{E}[c_{\text{final}} \mid b = \text{true}, c = c_0] \geq c_0 + \frac{1}{1-p}$, which matches the exact expected value of $c_0 + \mathbb{E}[T]$.

Canonical Witnesses: Worked Examples

The canonical witness sequence. Given a loop `while B do C` with invariant P , postcondition expectation g , and letting $q_n(m) = \Pr(T_m > n)$ denote the tail probabilities of the loop's hitting time T_m starting from state m , the canonical witness sequence is

$$r_n(m, c) \triangleq (c + n + 1) q_n(m) + \sum_{j=n+1}^{\infty} q_j(m). \quad (3)$$

This can equivalently be written as $r_n(m, c) = c \cdot q_n(m) + \mathbb{E}[T_m \cdot [T_m > n]]$, which makes plain why it decreases to zero (the truncated expectation vanishes as $n \rightarrow \infty$) and why it starts at g (the Abel/tail-sum identity $\sum_{n \geq 0} q_n(m) = \mathbb{E}[T_m]$ gives $r_0(m, c) = c + \mathbb{E}[T_m] = g(m, c)$).

We verify the four premises **P1–P4** of the WHILE-LB rule on two concrete programs.

Example 1: Geometric Termination

Program.

$$C_{\text{geo}} \triangleq \text{while } b \text{ } b := \text{Ber}(p); \ c := c + 1$$

The loop body samples a fresh Bernoulli(p) bit and increments the counter c . The loop terminates when $b = \text{false}$, which happens geometrically: $T_{b=\text{true}} \sim \text{Geom}(1-p)$, so $\mathbb{E}[T] = \frac{1}{1-p}$.

Postcondition and invariant. Take $P \equiv \mathbf{true}$ and

$$g(b, c) \triangleq c + \frac{[b]}{1-p}.$$

At the false-branch terminal state, $g(\mathbf{false}, c) = c$, which records the final value of c (number of loop iterations taken).

Tail probabilities. $q_n(\mathbf{true}) = p^n$ and $q_n(\mathbf{false}) = 0$.

Canonical witnesses.

$$r_n(\mathbf{true}, c) = (c + n + 1) p^n + \frac{p^{n+1}}{1-p},$$

$$r_n(\mathbf{false}, c) = 0.$$

Verification of P1. We need $\vdash_{\leq} \{ b \wedge P \mid g \} \rightsquigarrow_{C_{\text{body}}} \{ P \mid g \}$, i.e. $\text{wp}[b := \text{Ber}(p); c := c + 1](g) \geq g$.

$$\begin{aligned} \text{wp}[C_{\text{body}}](g)(b, c) &= p \cdot g(\mathbf{true}, c + 1) + (1 - p) \cdot g(\mathbf{false}, c + 1) \\ &= p \left(c + 1 + \frac{1}{1-p} \right) + (1 - p)(c + 1) \\ &= c + 1 + \frac{p}{1-p} = c + \frac{1}{1-p} = g(\mathbf{true}, c). \checkmark \end{aligned}$$

Equality holds, so P1 is satisfied.

Verification of P2. We need $r_0 \geq g$ on $B \wedge P$, i.e. at $b = \mathbf{true}$:

$$r_0(\mathbf{true}, c) = (c + 0 + 1) p^0 + \frac{p^1}{1-p} = c + 1 + \frac{p}{1-p} = c + \frac{1}{1-p} = g(\mathbf{true}, c). \checkmark$$

Verification of P3. We need $\text{wp}[C_{\text{body}}](r_n) \leq r_{n+1}$. Compute at $b = \mathbf{true}$:

$$\begin{aligned} \text{wp}[C_{\text{body}}](r_n)(\mathbf{true}, c) &= p \cdot r_n(\mathbf{true}, c + 1) + (1 - p) \cdot r_n(\mathbf{false}, c + 1) \\ &= p \left[(c + n + 2) p^n + \frac{p^{n+1}}{1-p} \right] + 0 \\ &= (c + n + 2) p^{n+1} + \frac{p^{n+2}}{1-p}. \end{aligned}$$

Meanwhile,

$$r_{n+1}(\mathbf{true}, c) = (c + n + 2) p^{n+1} + \frac{p^{n+2}}{1-p}.$$

These are equal, so P3 holds with equality. $\checkmark$

Verification of P4. On $B \wedge P$ (i.e. $b = \mathbf{true}$):

$$\lim_{n \rightarrow \infty} r_n(\mathbf{true}, c) = \lim_{n \rightarrow \infty} \left[(c + n + 1) p^n + \frac{p^{n+1}}{1-p} \right] = 0,$$

since $p \in (0, 1)$ so exponential decay dominates the polynomial prefactor. $\checkmark$

Example 2: Deterministic Countdown

Program.

$$C_{\text{det}} \triangleq \mathbf{while} \quad x > 0 \quad x := x - 1; \quad c := c + 1$$

The loop body decrements x and increments c deterministically. Starting from $x = k \geq 0$, the loop runs exactly k iterations: $T_k = k$.

Postcondition and invariant. Take $P \equiv \mathbf{true}$ and

$$g(x, c) \triangleq c + x.$$

Tail probabilities. Because $T_x = x$ deterministically, $q_n(x) = \Pr(T_x > n) = [x > n]$.

Canonical witnesses.

$$r_n(x, c) = (c + n + 1) [x > n] + \sum_{j=n+1}^{\infty} [x > j] = (c + n + 1) [x > n] + \max(x - n - 1, 0).$$

For $x > n$ this simplifies to $(c + n + 1) + (x - n - 1) = c + x$. For $x \leq n$ both terms vanish, giving $r_n(x, c) = 0$.
Hence

$$r_n(x, c) = (c + x) \cdot [x > n].$$

Verification of P1.

$$\text{wp}[C_{\text{body}}](g)(x, c) = g(x - 1, c + 1) = (c + 1) + (x - 1) = c + x = g(x, c). \checkmark$$

Verification of P2. At $x > 0$ (i.e. $B \wedge P$):

$$r_0(x, c) = (c + x) \cdot [x > 0] = c + x = g(x, c). \checkmark$$

Verification of P3.

$$\begin{aligned} \text{wp}[C_{\text{body}}](r_n)(x, c) &= r_n(x - 1, c + 1) \\ &= (c + 1 + (x - 1)) \cdot [x - 1 > n] \\ &= (c + x) \cdot [x > n + 1] \\ &= r_{n+1}(x, c). \checkmark \end{aligned}$$

Verification of P4. For any fixed x , $[x > n] = 0$ for all $n \geq x$, so $\lim_{n \rightarrow \infty} r_n(x, c) = 0$. $\checkmark$